

Home Bias and International Risk Sharing: Empirical Replication and Extension

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1 Introduction

The paper Sørensen, Wu, et al. (2007) argues that risk sharing and home bias are two related macroeconomic phenomena previously analyzed separately. While home bias refers to the observed tendency to hold disproportionately large shares of domestic assets (French and Poterba 1991, see also), risk sharing describes how well countries smooth consumption or income from idiosyncratic output shocks (Obstfeld 1994, see also). To analyze the potential link, the reference paper uses panel data for 24 OECD countries between 1993 and 2003. The results support their hypothesis regarding the close connection of both phenomena showing that a decreasing home bias is statistically significantly associated with increasing risk sharing – without claiming causality.

The rest of this report is organized as follows. Section 2 replicates the reference paper’s main results and assesses their robustness. Section 3 extends the time window and group sample. Section 4 then turns to the case study of the Euro Area (EA) to investigate whether the relationship of interest can be observed following the introduction of a common currency. This allows to analyze whether a decline in the exchange rate risk and transaction costs, two commonly argued reasons for home bias, are actually associated with a declining home bias and increasing risk sharing. Section 5 concludes.

2 Replication of Sørensen, Wu, et al. (2007)

For the sake of brevity, we are focussing our replication exercise on two main outputs. The first output plots (a scaled inverse of) cross-sectional risk-sharing estimates against the (natural logarithm of) foreign assets over GDP (Figures 1 and 2 in Sørensen, Wu, et al. (2007)). The second output runs Panel Regressions using a manually constructed (Equity) home bias measure explained subsequently (Tables 3 and 5 in Sørensen, Wu, et al. (2007)).

2.1 Equity home bias

Based on data from the External Wealth of Nations Database (EWN) Sørensen, Wu, et al. (2007) first report country-level foreign asset and liability holdings of equity, debt, and foreign direct investment relative to GDP for the 24 country sample. Using the same data from Lane and Milesi-Ferretti (2018) we are able to replicate their Table 1 precisely with only minor deviations presumably due to data correction (see Table A.1 in Appendix A.1).

home bias conceptually builds on a standard Capital Asset Pricing Model (CAPM) implying that all investors hold equal shares of risky assets. Adapted to the country level, a country i will hold a share of domestic equity or debt assets relative to the total portfolio equivalent to their world market share of equity or debt assets. However, empirically domestic assets normally account for a higher proportion in the portfolio resulting in international home bias. Potential causes for home bias named in the literature are currency risk (Adler and Dumas 1983), transaction costs (Cooper and Kaplanis 1994) as well as lack of information or market access. Sørensen, Wu, et al. (2007) use a standard measure in the literature for the home bias. The index goes from 0 to 1 where 0 is international diversification in line with the CAPM prediction and 1 is full home bias. More specifically, they compute:

$$\text{Equity home bias}_i = 1 - \frac{w_i^{\text{foreign}}}{1 - A_i} \quad (1)$$

where

$$w_i^{\text{foreign}} = \frac{\text{Foreign equity held by } i}{\text{Total equity portfolio}_i} ; \quad A_i = \frac{\text{Stock market cap}_i}{\text{World stock market cap}}$$

$$\text{Total equity portfolio}_i = \text{Stock market cap}_i + \text{Foreign equity held}_i - \text{Equity held by foreigners}_i$$

While the authors compute an equity home bias and a debt home bias using the latter for robustness checks, we limit our analysis on the Equity home bias (EHB) due to data constraints. Next to the EWN data (Lane and Milesi-Ferretti 2018) we use stock market capitalisation from the World Development Indicator Database¹ (World Bank 2026). Using their methodology we can replicate columns (1) and (2) of their Table 2, descriptively showing percentage of foreign equity in portfolio and the EHB. The results align closely (see Table A.2 in Appendix A.2). Small deviations could come from data updates or the deviating source for stock market capitalisation². The EHB exhibits a consistent decline across countries from 1993 to 2003.

Note that for the exploratory plots the original paper does not use the home bias measures but rather a rough proxy by looking at the natural logarithm of foreign assets over GDP. This proxy is inverse to the home bias, i.e. it increases over time. This measure is also used for robustness checks. As can be seen in Appendix A.1 Table A.1 and Figure 1 we recover their results in trend and level.

2.2 Cross-Section Risk Sharing and first replication result

For the cross-sectional risk-sharing regressions, we apply the methodology in Sørensen, Wu, et al. (2007, p.593) which coincides with the standard approach in the literature. Specifically, we run regressions of the form:

$$\Delta \log C_{it} - \Delta \log C_t = \text{constant} + \beta_{C,t}(\Delta \log \text{GDP}_{it} - \Delta \log \text{GDP}_t) + \varepsilon_{it}, \quad (2)$$

where C_{it} and GDP_{it} respectively measure per capita consumption and GDP of country i at year t . C_t and GDP_t denote the respective aggregates over the sample group at year t . Just as in Clymo (2026, p.21), we may interpret $\beta_{C,t} = 0$ as the perfect risk-sharing benchmark. In this case, idiosyncratic consumption growth perfectly tracks aggregate consumption growth and a country is fully insured against idiosyncratic output shocks.

For completeness, we may conclude with the following remark. While consumers ultimately derive utility from consumption, the authors also run 2 for GNI instead of consumption. In the context of this analysis, this may make sense because it relates more closely to the relationship under investigation: A decrease in the home bias manifests, *ceteris paribus*, in higher net foreign factor income and therefore insulates consumption by insulating (gross) income. Running 2 based on GNI may therefore provide a "better 'signal-to-noise' ratio" (Sørensen, Wu, et al. 2007, p. 589). It turns out that results are qualitatively similar which is why we focus on consumption risk sharing, and most often refer to the appendix for the replicated GNI risk sharing.

¹The paper uses Standard and Poor Global Stock Markets Factbooks which were not accessible.

²See Appendix A.3 for a brief description of key data cleaning steps.

Turning to results, Figure 1 Panel A reproduces Figure 2 of the reference paper which is displayed in Panel B for easy comparability. It becomes apparent that we are able to replicate the general trend proposed by Sørensen, Wu, et al. (2007): Both the share of Foreign Assets and the amount of Risk Sharing increased in the 1990s for the chosen selection of 24 OECD countries, inspiring the more detailed Panel Regressions described in the next section. Strikingly, the scaled cross-sectional risk-sharing estimates do not match the ones proposed in Sørensen, Wu, et al. (2007). There are two possible reasons for this. First, it may be that our underlying GDP-, GNI and consumption data may be different from the authors' – a possibility that we will discuss in the next section. Second, note that Sørensen, Wu, et al. (2007) only show smoothed risk-sharing estimates "by using a Normal kernel with bandwidth (standard deviation) equal to 2" (Sørensen, Wu, et al. 2007, p.594). While we tried to get as close as possible to this smoothing technique, it would have been helpful had the authors provided their original estimates. Plotting them for our data, we find, in the words of the reference paper, that the "year-by-year estimates fluctuate a fair amount" Sørensen, Wu, et al. (2007, p. 594), motivating the smoothing in the first place.

In conclusion, we regard the increasing trend in risk-sharing in the 1990s to be reliable, but are cautious about exact levels.

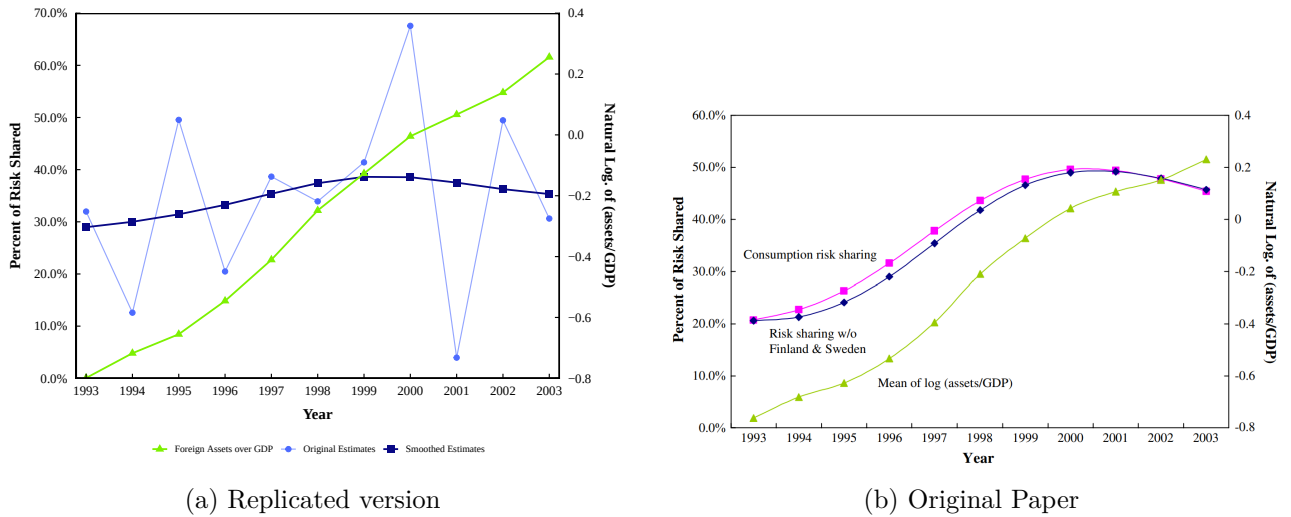


Figure 1: Consumption Risk Sharing and foreign asset holdings

2.3 Replication issues in Cross Section Risk Sharing

Throughout the process of our project, we were confronted with replicational difficulties regarding the cross-sectional regressions displayed on the left of Figure 1. While believe this partly reflects a sensitivity of the analysis to the data chosen. This section is devoted to this sensitivity.

The authors of the reference paper describe how they gather GDP, GNI and consumption data in only 8 lines in the appendix (Sørensen, Wu, et al. 2007, p.604). Relying on OECD data, they manually compute real aggregate variables using 1995 (consumer) prices. As the cross-section regression compares variables not only across time but also countries, they calculate PPP-adjusted values using Purchasing Power Parities provided by the OECD. Finally, they use population data to arrive at per capita values.

To show the sensitivity of the cross-sectional risk-sharing results, we replicated Figure 2 of Sørensen, Wu, et al. (2007) with three approaches, relying on OECD data for each of them. First, we download

GDP, GNI and consumption data from the OECD in current prices of the national currency. Using the country-specific CPIs, we convert to 2015 constant prices - the base year chosen by the OECD for most variables. We then use 2015 PPPs to arrive at PPP-adjusted data - all identical to the reference paper, with the exception of the base year having changed from 1995 to 2015. Approach 2 uses OECD data of all variables already PPP-adjusted using chained-linked volumes, rebased to 2020. Chained linked volumes are the modern way of estimating real aggregates and recommended by the OECD over choosing a constant base year. Finally, extracting data published on a PDF file by the OECD in 2005 using 2000 as a constant base year, Approach 3 most closely relates to Sørensen, Wu, et al. (2007), but with data that does not extend much the time period of the reference paper.

Approach 1 and the original version of the paper are in Panels A and B of Figure 1, while approaches 2 and 3 are Panels A and B of Figure 2.

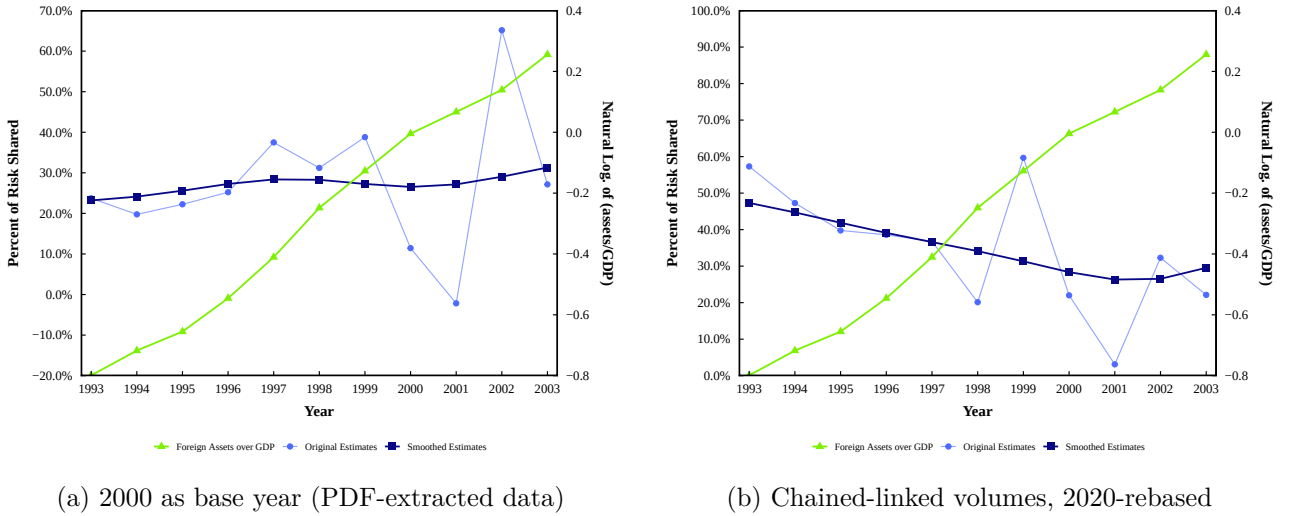


Figure 2: Consumption Risk Sharing (similar for GNI)

Most strikingly, using Approach 2, the trend of risk-sharing suddenly negative which would conclude the opposite of what the original paper claims. We believe this sensitivity to the data chosen is partly reflected by the fact that computing real aggregates (always amounting to choosing some base year) has to account for changes in underlying economic fundamentals. As discussed during the presentation, this task is almost certainly bound to fail which is why we conclude it to be worthwhile to get as close as possible to the base year chosen by the authors. Indeed, relying on 2000-constant prices gets us almost perfectly close for GNI cross-section risk-sharing. For consumption risk sharing (see 2), we also get closer. To be able to extend the framework proposed by Sørensen, Wu, et al. (2007), we stick with the data in 2015 prices, but alert to the sensitivity for future analysis.

2.4 Panel-Data Regressions

2.4.1 Theoretical outline

Sørensen, Wu, et al. (2007) use a panel regression framework to analyze whether a lower equity home bias is associated with a higher degree of income and consumption risk sharing. To do so, they regress the log growth rate of country-specific Gross National Income (GNI) or consumption (C), relative to the sample-wide log growth rate of GNI or consumption, on the country-specific log growth rate of

Gross Domestic Product (GDP), relative to the sample-wide log growth rate of GDP.³ The baseline regression is

$$\Delta \log C_{it} - \Delta \log C_t = \alpha_i + \kappa (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it}. \quad (3)$$

As in the cross-sectional regressions, one minus the respective coefficient κ measures the average degree of risk sharing.

To incorporate equity home bias into the panel regression, the authors allow the coefficient on relative GDP to vary with both a time trend and the deviation of a country's equity home bias from the sample average:

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(EHB_{it} - \overline{EHB}_t) \quad (4)$$

It is most intuitive to interpret the implied degree of consumption risk sharing CRS_{it} as

$$CRS_{it} = 1 - \kappa_0 - \kappa_1(t - \bar{t}) - \kappa_2(EHB_{it} - \overline{EHB}_t). \quad (5)$$

For the hypothesis of a potential co-movement between equity home bias and risk sharing, the main coefficient of interest is $-\kappa_2$. A positive value of $-\kappa_2$ implies that a higher equity home bias increases the sensitivity of relative consumption growth to relative GDP growth and therefore reduces consumption risk sharing. Equivalently, an increase in equity home bias by 0.1 changes consumption risk sharing by $-0.1\kappa_2$. All panel regressions are estimated using a two-stage feasible GLS-type approach.⁴

2.4.2 Empirical results

When reporting the results of the panel regressions, we follow the authors standard and report the easier to interpret coefficients $1 - \kappa_0$, $-\kappa_1$ and $-\kappa_2$ multiplied by 100 to interpret them as percentage effects. Our panel regression results follow the key results of Sørensen, Wu, et al. (2007), with a statistically non-different to zero estimate for the time trend and a highly significant coefficient for the equity home bias both on income and consumption risk sharing. An increase in the equity home bias by 0.1 is now associated with an increase in income risk sharing by roughly 13 percent and consumption risk sharing by roughly 20 percent. Given that our point estimates for the coefficient are larger than 100 and notably higher than the point estimates found by Sørensen, Wu, et al. (2007),⁵ we rely on an interpretation offered by the authors themselves, believing that a coefficient of more than 100 percent risk sharing likely indicates a strong co-movement, even if the size of the point estimate is not likely to hold in larger samples. A statement that proves to be true once we extend the sample period from 1993–2003 to 1987–2017. Our coefficient for average income risk sharing is highly significant, while it is not significant in the original paper. Given that it will turn non-significant in the larger sample period, we observe this, but doubt that it affects the main findings of this exercise.

Running the same regressions on foreign asset holdings over GDP, as done by Sørensen, Wu, et al.

³See Sørensen, Wu, et al. (2007) for a more extensive discussion. For the sake of simplicity we decided to only show the relevant theoretical specification with consumption. In the tables below, we report the regression both for GNI and for consumption.

⁴In the second-stage regression, each country is assigned a weight proportional to one over the standard error of its first-stage residuals. Since this procedure is weakly documented in the original paper, we compared the first- and second-stage regression results. The differences do not affect the direction or statistical significance of the main coefficients.

⁵we believe that the discussed data choices and data revisions and the choice of the base year play an important role.

Table 1: Income and Consumption Risk Sharing and Equity Home Bias: OECD 1993–2003

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	16 (4.16)	1 (0.76)	−133 (4.87)
Consumption	55 (15.10)	1 (0.52)	−204 (9.07)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. The income and consumption row present $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

(2007) confirms the robustness of the replication. Even though we find partly differing point estimates, the direction and significance of the point estimates are in line with the original paper.⁶ We treat these regressions as an robustness check and find positive $-\kappa_2$ coefficients for all forms of asset holdings, and statistically non significant estimates for the time trend. The results are reported in Table A.3 of Appendix B.2.

Summarizing, we conclude that the main findings of Sørensen, Wu, et al. (2007) panel regressions can be replicated with updated data.

3 Extensions by Time and Sample Size

As a logical extension and robustness check of the results in Sørensen, Wu, et al. (2007) we apply the entire methodology to a longer time sample with the original countries. We tried to extend the sample to more countries either. Due to limitations on both sides, the EWN and WDI data as well as the OECD data, the resulting sample was barely larger than before. Results and details on data limitations are thus reported in Appendix B.2.

The considered time horizon was chosen based on the home bias data whichs methodology remains the same. Due to limited data coverage before the 1980s, we decided to start the series in 1987, five years before the Maastricht Treaty. In the second half of the 2010s, the WDI world market capitalization is less complete, especially for EU countries. Hence, we limited the long sample to 2017 also considering potential irregular patterns during Covid-19 and Ukraine crises.

Looking at figure (XXX), we can see that the cross section risk sharing coefficient has increased constantly from 1987 to 2000, using the same smoothing method as the authors in the paper. After a short dip during 2000 to 2003, potentially caused by the *dotcom*-bubble, we observe an even stronger increase in cross section risk sharing that is ended abruptly by the global financial crisis 2007/2008. After this crisis, global cross section risk sharing grows even stronger. Looking at the curve for foreign assets over GDP we can see a constant increase over the time period of 31 years.

By extending the time horizon for the original sample of countries, we can confirm that Sørensen, Wu, et al. (2007) findings are robust on a longer time horizon, which is an important finding given the rather short original sample period.

Panel-data regressions find that the main relationship between equity home bias and risk sharing largely hold when extending the time period by 20 years. A 0.1 decrease of the equity home bias is now associated with a 3 % increase in income risk sharing and a 8 % increase in consumption risk sharing. The average risk sharing coefficient decreases notably, and becomes statistically non-significant for

⁶Note that a positive sign of $-\kappa_2$ is now associated with a higher level of risk sharing.

income risk sharing.⁷ Most relevant is the significance of $-\kappa_2$. / Focusing on foreign asset over GDP the regression Table B.1 in Appendix B.1 adds further robustness and lets us conclude that the negative relationship between home bias and risk sharing holds over the extended time period.

Table 2: Income and Consumption Risk Sharing and Equity Home Bias: OECD 1987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.14)	0 (0.82)	−28 (2.80)
Consumption	40 (19.61)	0 (0.29)	−81 (7.96)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. Extended time period 1987–2017 For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

The slightly extended sample across countries and time, leads to further robust results shown in Appendix B.3.

4 Case Study Euro Area

As a central extension, we examine the EA as a case study. The introduction of the euro eliminated exchange rate risk among member states and, fueled by further European Integration, transaction costs for cross border financial flows fell. As introduced in Section 2.1 these are commonly seen as drivers of home bias. The EA therefore provides a setting where we would expect a pronounced decline in home bias compared to similar countries. Following the established association risk sharing should increase correspondingly.

4.1 Case Study Euro Area: Panel Regression

Using the EA as its own sample leads to rather non-intuitive results, both on a cross sectional and panel regression level. As can be seen in Table 3, the point estimates are notably smaller for the EA than for the OECD sample, and the coefficient for the equity home bias on consumption risk sharing turns non significant. The same effect can be seen in the robustness check using foreign assets over GDP in Appendix C.1 Table C.1. Sørensen, Wu, et al. (2007) note the same puzzling irregularities when looking at an EU sample of 13 EU member states, especially with respect to the consumption risk sharing coefficient among EU countries. They cite Balli and Sørensen (2006) and speculate that consumption risk sharing within the EU has decreased due to less-pro cyclical government spending among EU countries following the introduction of the Maastricht criteria. We understand that this reasoning can also be applied to the EA and tried to run structural break regressions to analyse this relationship further. Due to partly significant but non robust results in those structural break regressions, we investigate both components separately in the following.

⁷Given that this coefficient was non significant in Sørensen, Wu, et al. (2007) for the original sample period, we conclude that it its significance varies over time and data availability and not too much weight should be given on its interpretation.

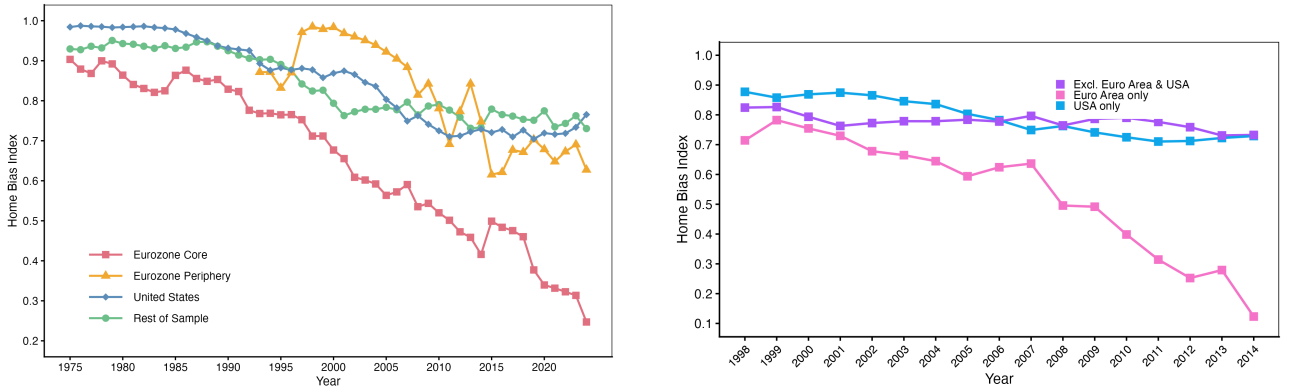
Table 3: Income and Consumption Risk Sharing and Equity Home Bias: Euroarea 1987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.09)	0 (0.96)	−30 (2.16)
Consumption	27 (8.85)	−1 (3.48)	−8 (0.50)

Note: Values in parentheses are absolute t-values. Euroarea sample for the period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

4.2 Case Study Euro Area: Equity home bias

To simplify the joint analysis and find potential reasons for the surprising results we start with the home bias in the EA. As argued before, we expect a stronger home bias decrease of EA countries than other developed economies after the introduction of the euro. Following this logic, the home bias of the EA as an aggregate entity should decrease to a much lesser extent if asset holding diversification mainly occurs within the EA. Surprisingly, this is not in line with the EHB based on EWN data we observe. Indeed, both, the cross country average weighted by total equity portfolio and the EA-aggregate home bias decrease as can be seen in Figure 3a and Figure 3b. But the decline of the aggregate group is stronger in absolute terms and in comparison to other developed economies or the US. This would imply that EA portfolios became more diversified, but most diversification occurred outside the EA.



(a) Total equity weighted mean by country-group

(b) Euro area aggregate vs. full sample and US

Figure 3: Equity home bias in the euro area and other developed economies

Note: Eurozone Core = founding members; Eurozone Periphery = joined 2001–2015.

Based on this result at odds with theory and the in general suspiciously strong decline in EA-aggregate EHB approaching 0, we further investigated related research.

4.2.1 Financial Centers in Europe

An interesting explanation can be found in Beck et al. (March 2024). This ECB working paper explores the role of Financial Centers in the EA and how they distort foreign equity positions and home bias measures. Ireland, Luxembourg and (lesser) the Netherlands host large finance industries and more specifically investment funds and centers for securities issuance for foreign firms due to regulatory and tax reasons. Standard international investment positions (IIP) like those reported in

the EWN database only consider the country of residency of a) the holder and of b) the fund or financing subsidiary but ignore the location of the underlying assets. Additionally, fund shares are classified as equity under BPM6⁸ regardless of underlying assets. This leads to various distortions in the data and EHB measures: (i) If EA countries hold EA fund shares with non-domestic EA assets underlying, there is no distortion because the equity is labeled foreign which is all that matter for the national EHB, and the misattribution cancels out in the EA aggregate. (ii) If EA countries hold EA fund shares with domestic assets underlying (round-tripping) the foreign equity position is overstated leading to a too low EHB measure. (iii) If Euro Area countries hold EA fund shares with non EA assets underlying, the national EHB is not affected neither is the EA because the effects on foreign equity portfolio cancel out. (iv) If non-EA countries hold EA fund shares with EA assets underlying, there is no distortion to the non-EA country's EHB. The EHB of the fund domiciled country is downward biased because its foreign equity portfolio is higher by intermediating the extra EAs position to the owner country of the underlying asset. For the EA aggregate these effects cancel out. (v) If non EA countries hold EA fund shares with non EA assets underlying, the foreign equity portfolio of Ireland or Luxembourg is overstated and their EHB is too low. The same is true for the EA aggregate EHB. (vi) If debt assets are underlying the fund the foreign equity portfolio is overstated. Subsequently, the foreign equity share is inflated and the EHB distorted downwards.

4.2.2 Data Analysis

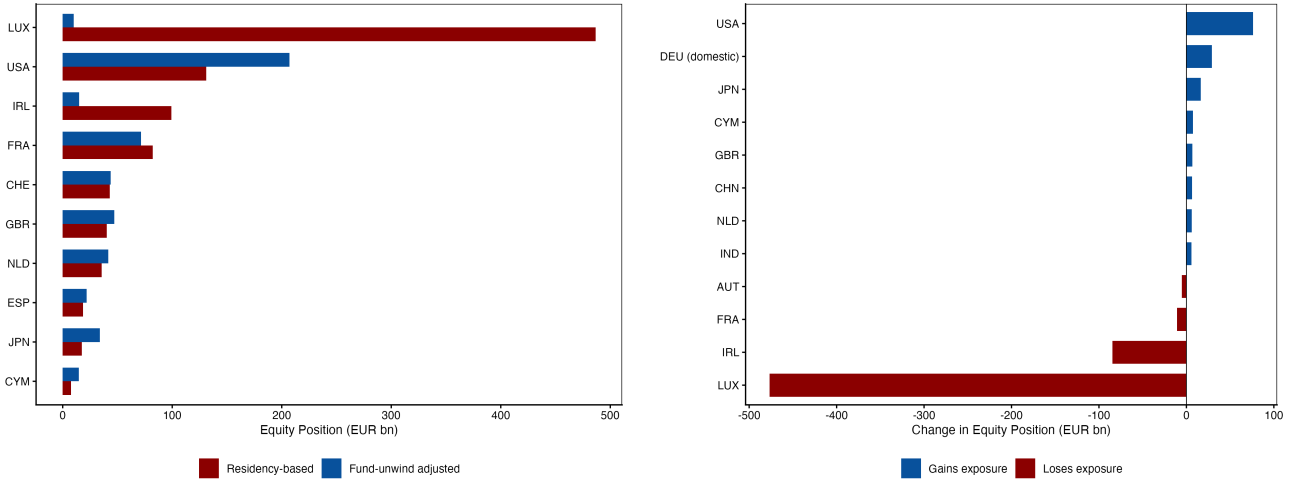
To get a better impression of the misattribution and the true relations underneath the surface we explore the unwind data provided by Beck et al. (March 2024). First, it is important to note that the volume traded in these funds and thus the magnitude of the underlying problem strongly increased over time. In 1998 the volume of Luxembourg domiciled funds was below € 500 billion increasing to below € 1000 billion in 2003. By 2014 the volume already increased to over € 2500 billion, reaching more than € 5.500 billion in 2021. The trend is very similar for Ireland. See Figure 9a and Figure ?? in Appendix C.5 for a graphical analysis. This implies that the EHB data used for the original paper is only partly affected. Ireland was excluded due to data problems and Luxembourg is not in the sample. Hence, the problem is restricted to round tripping of other countries via these funds. While the two countries are still excluded from the time extended sample on the one hand, the distortion of other countries EHB increased with the volume of the funds⁹ and on the other hand the EA observation is affected, as discussed before.

Secondly, we look at Germany as an example to illustrate the dynamics. Figure 4a shows the IIP of Germany vis a vis other countries before (red) and after (blue) the data correction. While the position increases against most other countries, it drops dramatically for Ireland and especially Luxembourg. Equity that was attributed to Luxembourg is now correctly attributed to e.g. the US. Figure 4b displays the net change of Germany's investment positions (corrected data minus standard data). Next to the before discussed pattern, this shows how the foreign equity holding for Germany is overstated if the underlying securities are themselves German.

Due to missing market capitalisation data in the WDI database, we were not able to compute corrected EHB values. Furthermore the corrected data is only available since 2014 making a corrected panel regression impossible. For an additional analysis of the foreign equity portfolio share, one of the two components of the EHB, see Appendix C.5. While interesting in itself, the analysis provided little

⁸Balance of Payments and International Investment Position Manual, 6th edition (International Monetary Fund 2009).

⁹assuming a slower growth of other equity positions



(a) Residency-based vs. fund-unwind adjusted

(b) Net change after fund-unwind correction

Figure 4: Germany's top foreign equity positions (2018q4), Source:Beck et al. (March 2024)

insight in the trajectory of the EHB for EA countries and EA aggregate after the Euro introduction.

4.2.3 Home bias Trajectory

To look at the true development of the home bias we turn to results from Beck et al. (March 2024).

Their EHB results with the raw data using the same formula shows a similar pattern for EA countries and the US as in Figure 3a. However, after correction of the data, the cross country average exhibits a very similar decline as the US. Nonetheless, for the Debt home bias (DHB) even after correcting the data, the bias strongly declines around the introduction of the Euro and stabilises at a significantly lower level than the US. The paper does not report the corrections for the EA aggregate leaving the question open how much of the (debt) home bias decline came from EA integration or from a general trend for internationalisation of asset holdings. Further analysis is necessary to understand the dynamics in the home bias after the EA creation and the panel regression results must be interpreted cautiously.

4.3 Case Study Euro Area: Risk Sharing

The previously described data issues on EWN data notwithstanding, it might still be worthwhile to analyze how risk-sharing developed in the Euro Area. We devote our final section to this analysis. For a first preliminary indication, consider Figure 5. It displays the development of consumption risk sharing, using either Euro Area countries (Panel A) or non-Euro Area countries only (Panel B). Importantly, we depart from the baseline methodology by choosing the same aggregate for both subgroups. As discussed in the presentation, using the same aggregate allows us to compare the shock insulation of EA countries with that of non EA countries using the same baseline shock¹⁰. While it becomes apparent that between the introduction of the Euro in 1999 and the eve of the Great Financial Crisis (GFC) in 2007, the increase in the consumption risk sharing is more pronounced in the Euro Area sample, the difference is comparatively low (30% vs. 25% increase). Furthermore, as explained in the first section, we argued that we regard the exact levels of risk sharing to be base-year-sensitive in any case. Hence, we may conclude that consumption risk sharing has not markedly increased more

¹⁰However, results do not differ much when using the within-group aggregate, see Figure 11

in the Euro Area than in other countries.

To interpret this finding, recall our hypothesis based on Sørensen, Wu, et al. (2007): the Euro's 1999

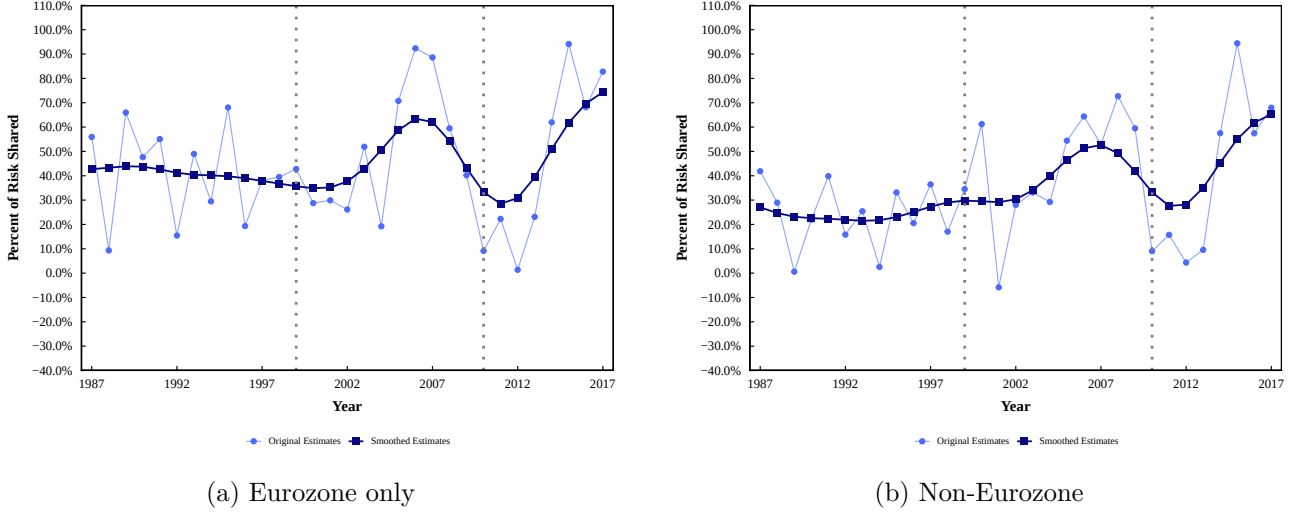


Figure 5: Consumption Risk Sharing cross-sectional estimates (same aggregate)

introduction eliminated currency risks, which *may* reduce the home bias and increase risk sharing, *ceteris paribus*. However, Euro Area home bias did not fall more than in other countries, which is consistent with consumption risk sharing similarly lacked disproportionate growth. However, because concurrent changes in other factors (e.g., federal transfers, borrowing) likely violate the *ceteris paribus* assumption, we must specifically isolate the contribution of capital markets to risk sharing to make this statement more robust. Reassuringly, Asdrubali, Sorensen, and Yosha (1996) develop a convincing method to isolate different channels of risk sharing. It builds on the accounting differences between GDP, GNI, Gross National Disposable Income (GDI) and Consumption using:

$$\text{GDP}_{it} = \frac{\text{GDP}_{it}}{\text{GNI}_{it}} \frac{\text{GNI}_{it}}{\text{GDI}_{it}} \frac{\text{GDI}_{it}}{C_{it}} C_{it} \quad (6)$$

Taking logs and differencing, one arrives at:

$$\begin{aligned} \Delta \log \text{GDP}_{it} = & (\Delta \log \text{GDP}_{it} - \Delta \log \text{GNI}_{it}) + (\Delta \log \text{GNI}_{it} - \Delta \log \text{GDI}_{it}) + \\ & (\Delta \log \text{GDI}_{it} - \Delta \log C_{it}) + \Delta \log C_{it} \end{aligned} \quad (7)$$

Proceed by evaluating $(\mathbf{XX})$ at the cross-sectional mean and subtracting it from $(\mathbf{XX})$. Multiplying with $(\Delta \log \text{GDP}_{it} - \Delta \log \overline{\text{GDP}}_t)$ and evaluating in expectation gives:

$$\begin{aligned} \text{Var}(\Delta \log \text{GDP}_{it}) = & \text{Cov}(\Delta \log \text{GDP}_{it} - \Delta \log \text{GNI}_{it}, \Delta \log \text{GDP}_{it}) + \\ & \text{Cov}(\Delta \log \text{GNI}_{it} - \Delta \log \text{GDI}_{it}, \Delta \log \text{GDP}_{it}) + \\ & \text{Cov}(\Delta \log \text{GDI}_{it} - \Delta \log C_{it}, \Delta \log \text{GDP}_{it}) + \text{Cov}(\Delta \log C_{it}, \Delta \log \text{GDP}_{it}) \end{aligned} \quad (8)$$

Finally, we may divide by the left-hand side to arrive at an equation including four simple OLS-coefficients that, by construction, sum to one:

$$\beta_C + \beta_F + \beta_C + \beta_U = 1. \quad (9)$$

Consider just the first coefficient. It is captured by the OLS-regression $\Delta \log \text{GDP}_{it} - \Delta \log \text{GNI}_{it} = \alpha_C + \beta_C \Delta \log \text{GDP}_{it} + \epsilon_{it}$ and its coefficient therefore measures the degree to which idiosyncratic output growth can explain the growth differential between GDP and GNI. As the difference between GDP and GNI relates to net factor income from abroad, Asdrubali, Sorensen, and Yosha (1996) interpret β_C as the amount of risk-sharing obtained via cross-ownership of assets, i.e. capital markets. Similarly, we may interpret β_F , β_C and β_U as the amount of risk sharing obtained via federal transfers, credit markets (borrowing/ lending) and an unsmoothed residual, respectively.

To apply the method, we first gather the same data as the well-documented working paper Hosseini and Gervais (March 2026). We are able to replicate both Table 7 and Figure 4, which runs Panel and Cross-Section Regressions for an EU27 sample and use this to reassure us of the correct methodological implementation. Then, we restrict ourselves to the Euro Area and construct year-by-year estimates of different risk sharing channels in the Euro Area, plotted in Figure 6, constructed equivalently to Hosseini and Gervais (March 2026, p.23).

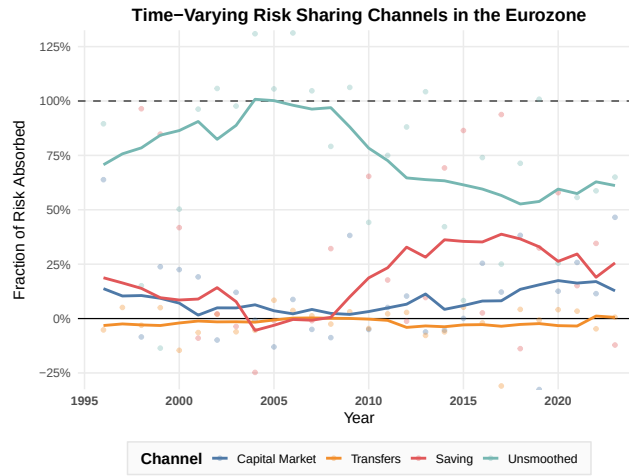


Figure 6: Year-by-year cross-sectional estimates (Smoothed with 10-year centred moving averages)

Graphically, it becomes evident that the capital markets channel behaves almost flat, though it increases markedly after the GFC. Indeed, relying on Panel Regressions, it is insignificant for the 1995-2007 window, and turns significant using 2010-2023 (see Figure). Hence, the mere abolishment of exchange-rate risk most likely did not reduce the home bias enough to disproportionately increase consumption risk sharing in the Euro Area. We therefore confirm the previous finding: Contrary to our initial hypothesis, the elimination of currency risks alone did not increase risk-sharing by much. This finding complements existing literature nicely. Sørensen and Yosha (1998, p.214) state that "if [...] cost[s] of transacting in many currencies have been the main cause of little cross-country ownership, their removal may induce a swift process of capital market integration." Evidently, our analysis shows that capital market integration, if significant, has neither been sizeable nor swift. Second, as hypothesized already in our reference paper, "hedging of currency risk is not the main reason for home bias" (Sørensen, Wu, et al. 2007, p.591). Our extensions are consistent with this statement.

5 Conclusion

This report looked at the relationship between the tendency to hold a disproportionate amount of domestic assets (the *home bias*) and risk sharing, relying on Sørensen, Wu, et al. (2007). We were

able to replicate the reference paper's main finding that the two phenomena are negatively correlated: A decrease in home bias is associated with an increase in risk sharing due to diversification of factor income streams. While we showed the finding to be robust to time and group extensions, we cast some doubt on the specific levels of risk sharing and raised the issue of choosing a base year as close as possible to a preferably small sample window.

We then focussed our analysis on the Euro Area, initially hypothesizing that the introduction of a common currency would reduce the home bias and increase risk sharing by more than in non-member states. Analyzing further the underlying home bias data, we find that the (Equity) home bias did not fall markedly after 1999. Consistent with this finding, we were also able to confirm that risk sharing did not disproportionately increase.

This counterintuitive finding reinforces a common policy suggestion: The Euro Area should make it a priority to increase risk sharing through other measures to get closer to US interstate levels (a successful monetary union) where only around 20% of output shocks are unsmoothed and capital markets smooth more than 50% of consumption. Otherwise, the cost of forming a monetary union (a common monetary policy) are particularly large.

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Appendices

A Appendix A: Replication

A.1 Appendix A: Paper Table 1 Replication

Table A.1: County-level foreign asset and liability holdings of equity, debt, and foreign direct investment relative to GDP

Country	Assets						Liabilities					
	Equity		Debt		FDI		Equity		Debt		FDI	
	1993	2003	1993	2003	1993	2003	1993	2003	1993	2003	1993	2003
Australia	0.07	0.17	0.08	0.18	0.15	0.34	0.12	0.26	0.33	0.49	0.28	0.40
Austria	0.02	0.17	0.47	1.26	0.04	0.22	0.02	0.10	0.31	0.98	0.06	0.22
Belgium	0.27	0.46	1.48	2.42	0.29	0.99	0.04	0.09	0.39	0.62	0.43	1.12
Canada	0.12	0.31	0.17	0.22	0.26	0.59	0.07	0.19	0.47	0.39	0.33	0.52
Denmark	0.06	0.24	0.54	0.81	0.14	0.47	0.02	0.14	0.56	0.59	0.13	0.46
Finland	0.00	0.21	0.29	0.80	0.14	0.54	0.06	0.60	0.55	0.56	0.09	0.39
France	0.04	0.19	0.50	1.05	0.21	0.41	0.08	0.27	0.20	0.62	0.23	0.29
Germany	0.06	0.23	0.45	1.07	0.08	0.35	0.05	0.14	0.19	0.54	0.03	0.33
Greece	0.00	0.02	0.18	0.47	0.01	0.07	0.01	0.08	NA	0.60	0.09	0.12
Iceland	0.00	0.28	0.04	0.36	0.01	0.18	0.00	0.03	0.25	0.89	0.01	0.13
Ireland	0.26	1.35	0.79	6.42	0.56	1.17	0.32	2.93	NA	1.23	0.66	2.08
Italy	0.03	0.21	0.29	0.59	0.08	0.16	0.04	0.14	0.14	0.55	0.05	0.15
Japan	0.02	0.06	0.38	0.51	0.06	0.08	0.04	0.12	0.08	0.07	0.00	0.02
Mexico	0.00	0.01	0.05	0.08	0.02	0.06	0.09	0.07	NA	0.11	0.08	0.21
Netherlands	0.17	0.61	0.77	1.94	0.34	3.73	0.26	0.56	0.20	1.64	0.21	2.95
New Zealand	0.01	0.22	0.05	0.24	0.22	0.25	0.08	0.11	0.33	0.45	0.47	0.57
Norway	0.04	0.33	0.21	0.96	0.10	0.36	0.08	0.10	0.11	0.36	0.11	0.21
Portugal	0.01	0.08	0.27	1.27	0.02	0.30	0.03	0.23	0.06	0.55	0.16	0.48
Spain	0.01	0.09	0.27	0.73	0.05	0.27	0.04	0.21	0.15	0.38	0.15	0.49
Sweden	0.07	0.43	0.26	0.54	0.21	0.77	0.10	0.27	0.12	0.57	0.06	0.67
Switzerland	0.33	0.81	1.57	2.85	0.36	1.07	0.51	1.11	0.08	0.13	0.20	0.63
Turkey	0.00	0.00	0.08	0.12	0.00	0.02	0.01	0.03	0.05	0.07	0.04	0.10
United Kingdom	0.25	0.37	1.23	2.12	0.22	0.76	0.17	0.53	0.22	0.51	0.17	0.43
United States	0.08	0.18	0.19	0.29	0.18	0.27	0.05	0.16	0.15	0.32	0.13	0.24

A.2 Appendix A: Paper Table 2 Replication

Table A.2: Equity Home Bias 1993 and 2003

Country	1993		2003	
	Foreign equity in portfolio (%)	Equity Home Bias	Foreign equity in portfolio (%)	Equity Home Bias
Australia	11.33	0.89	17.23	0.82
Austria	14.12	0.86	59.50	0.40
Belgium	45.97	0.54	50.17	0.50
Canada	11.16	0.88	27.21	0.72
Denmark	17.85	0.82	36.57	0.63
Finland	1.66	0.98	NA	NA
France	12.86	0.87	28.23	0.70
Germany	24.87	0.74	44.57	0.54
Greece	NA	NA	4.62	0.95
Iceland	NA	NA	NA	NA
Ireland	NA	NA	NA	NA
Italy	22.07	0.78	45.70	0.53
Japan	3.82	0.95	10.29	0.89
Mexico	1.16	0.99	12.48	0.87
Netherlands	40.03	0.59	68.53	0.30
New Zealand	2.80	0.97	42.60	0.57
Norway	22.23	0.78	50.99	0.49
Portugal	NA	NA	37.38	0.63
Spain	5.85	0.94	13.05	0.87
Sweden	16.05	0.84	41.60	0.58
Switzerland	40.28	0.59	47.51	0.51
Turkey	1.83	0.98	2.33	0.98
United Kingdom	23.18	0.75	36.17	0.61
United States	10.03	0.84	14.33	0.73
Average	16.46	0.83	32.91	0.66

A.3 Appendix A: Data Processing Description

Next to some missing data points leading to missing EHB values, e.g. for Ireland, we excluded some observations. First, in few cases the total equity portfolio was negative leading to EHB values larger 1. Secondly, in even less cases EHB was negative which would be equivalent to a foreign bias. Both problems are most probably due to the data mismatch combining the EWN and WDI data. Problems occurred for Finland from 1999 to 2003 potentially still related to the Finnish Banking crisis (Honkapohja and Koskela 1999) and in Italy, Austria, and France for 1998, where each countries stock market capitalisation rapidly dropped before going back to the old level the next year. In 1998, both the Italian and the Austrian stock exchange were restructured or privatised potentially causing these data discontinuities (Wiener Börse 2026; Borsa Italiana 2026). Ireland had both, years with negative total equity or negative EHB. We completely discarded Ireland from the analysis. Potential explanations for this will be discussed in Section ??.

A.4 Appendix A: GNI Exploratory Cross Section Plot

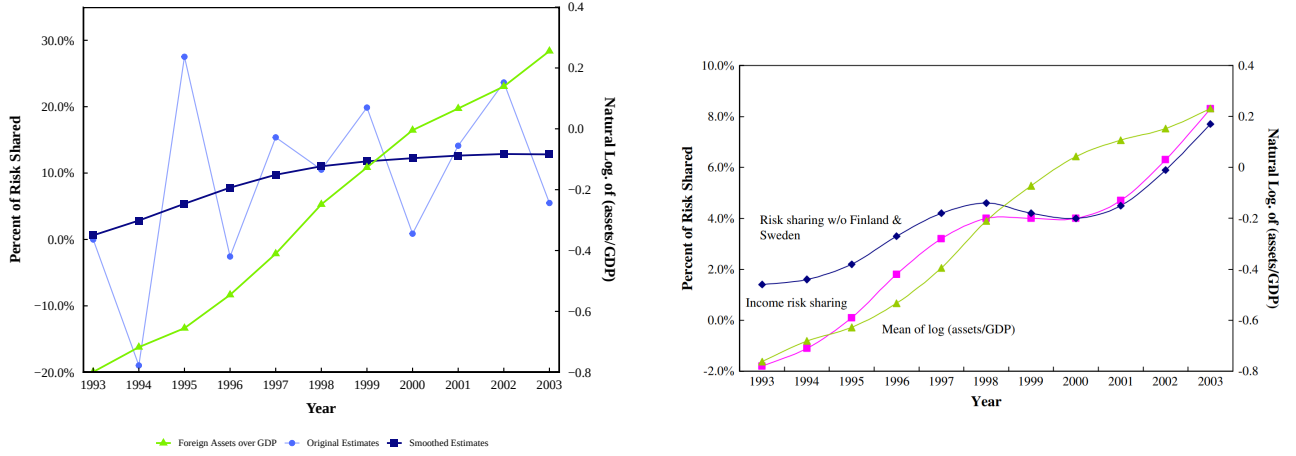


Figure 7: Income Risk Sharing and foreign asset holdings

A.5 Appendix A: Cross Section Regression Results Assets/GDP

Table A.3: Risk Sharing and Foreign Asset Holdings Relative to GDP: OECD 1993–2003

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	15 (4.15)	1 (1.21)	12 (5.19)				
	16 (4.21)	1 (0.72)		18 (5.15)			
	10 (2.79)	2 (1.63)			12 (3.33)		
	16 (4.25)	1 (0.73)				18 (5.05)	
	16 (4.26)	1 (1.06)					21 (5.14)
Consumption risk sharing	57 (14.77)	1 (0.90)	18 (8.80)				
	50 (12.16)	0 (0.15)		21 (6.01)			
	52 (12.59)	0 (0.41)			14 (6.16)		
	52 (12.93)	0 (0.32)				23 (6.81)	
	55 (13.59)	0 (0.46)					24 (7.35)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

B Appendix B: Within Paper Extension

B.1 Appendix B: Long Sample Cross Section Regression Assets/GDP

Table B.4: Risk Sharing and Foreign Asset Holdings Relative to GDP: OECD 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	−2 (0.87)	0 (0.70)	5 (3.59)				
	0 (0.21)	0 (0.80)		8 (4.14)			
	−2 (0.98)	0 (0.40)			4 (2.57)		
	−1 (0.62)	0 (0.91)				8 (4.15)	
	−2 (1.03)	0 (0.74)					8 (4.08)
Consumption risk sharing	42 (20.31)	0 (0.28)	10 (9.07)				
	39 (18.10)	0 (0.72)		12 (5.60)			
	40 (19.21)	0 (0.18)			11 (7.41)		
	40 (18.98)	0 (0.58)				14 (7.06)	
	41 (19.35)	0 (0.40)					15 (7.60)

Note: Values in parentheses are absolute t-values. Original sample and extended time period. For income risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. For consumption risk sharing, the reported coefficients are $100 \cdot (1 - \eta_0)$, $100 \cdot (-\eta_1)$, and $100 \cdot (-\eta_2)$. Country-fixed effects included.

B.2 Appendix B: Broad Sample Construction

For the broader country sample to address the coverage we first limited the data to the joint subset of the top 50 economies based on the EWN reported GDP values combined with all EU27 countries. Starting from there we analysed irregularities in the EHB index as before (excluding e.g. Luxembourg and Ireland due to negative total equity portfolio values). Because the regression uses deviations from the mean we tried to avoid frequent drop ins or outs. Instead of keeping countries with very few non NA values for EHB, we eventually excluded Ireland, Luxembourg, Finland, Malta, Cyprus, Lithuania, Latvia and Estonia as well as Taiwan from the broader sample. All remaining countries in the broad sample have an EHB coverage of at least 40%. For a visual analysis of the data coverage after excluding irregular observations see Figure 8.

Merging the sample with the OECD data only 29 countries remain when excluding all countries with missing GNI, GDP, consumption or population data.

B.3 Appendix B: Broad Sample Cross Section Regression

Extending the OECD sample to 29 countries ¹¹ for the sample period 1987–2017 gives us the opportunity to analyse if the main findings of Sørensen, Wu, et al. (2007) can be extended to a broader set of countries. As a potential shortcoming of this extension, we must admit that the sample size is still rather small. The panel-regression results of the (slightly) extended sample between 1987 and 2017 confirm the same results as shown in the two regressions before. An decrease in the Equity home bias by 0.1 is associated with an increase of income risk sharing by roughly 3 percent and an increase in consumption risk sharing by roughly 8 percent.

Again, the panel regressions for foreign Asset holdings relative to GDP confirm this estimation, with highly significant estimations for the asset holdings themselves, and non significant estimations for the time trend. Again, the average risk sharing coefficient for income is non significant in both specifications, while it is highly significant for consumption in both specifications.

Table B.5: Income and Consumption Risk Sharing and Equity Home Bias: Extended OECD Sample 19987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.34)	0 (0.26)	–28 (3.24)
Consumption	30 (15.28)	0 (0.41)	–84 (8.25)

Note: Values in parentheses are absolute t-values. Extended sample with up to 29 OECD countries, many observations lost due to missing data coverage. Extended time period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

¹¹For the GNI analysis and panel regressions some observations drop, due to a lack of data coverage for both Consumption, GNI, GDP and the Equity home bias. A future extension of Sørensen, Wu, et al. (2007) should certainly try extend the country sample beyond OECD countries. Due to our data problems, we decided to keep a rather small, but consistent sample.

Table B.6: Risk Sharing and Foreign Asset Holdings Relative to GDP: Extended OECD Sample and 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	1 (0.63)	0 (0.68)	3 (3.40)				
	0 (0.20)	0 (0.93)		7 (4.72)			
	−2 (1.23)	0 (0.62)			4 (3.23)		
	0 (0.05)	0 (1.09)				7 (4.65)	
	0 (0.27)	0 (0.94)					7 (4.54)
Consumption risk sharing	28 (14.42)	0 (0.56)	12 (11.26)				
	30 (14.02)	0 (0.49)		10 (4.75)			
	26 (11.85)	0 (2.00)			12 (7.78)		
	28 (13.53)	0 (0.00)				13 (6.73)	
	28 (13.46)	0 (0.27)					14 (7.37)

Note: Values in parentheses are absolute t-values. Extended sample with up to 29 OECD countries, many observations lost due to missing data coverage. Extended time period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

C Appendix C: Eurozone Case Study

C.1 Appendix C: Eurozone Cross Section Regression Results Assets/GDP

Table C.7: Risk Sharing and Foreign Asset Holdings Relative to GDP: Euroarea 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	1 (0.15)	0 (0.17)	3 (0.86)				
	−7 (1.15)	0 (0.77)		11 (2.11)			
	−4 (0.79)	0 (0.22)			7 (1.76)		
	−6 (1.02)	0 (0.64)				11 (2.00)	
	−7 (1.25)	0 (0.55)					12 (2.35)
Consumption risk sharing	24 (6.21)	−1 (2.99)	−3 (1.36)				
	27 (8.79)	−1 (3.37)		0 (0.03)			
	27 (9.04)	−1 (2.72)			6 (1.67)		
	27 (8.88)	−1 (3.41)				0 (0.01)	
	27 (8.93)	−1 (3.45)					4 (0.81)

Note: Values in parentheses are absolute t-values. Euroarea sample for the period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

C.2 Appendix C: Luxembourg and Ireland Fund Ownership

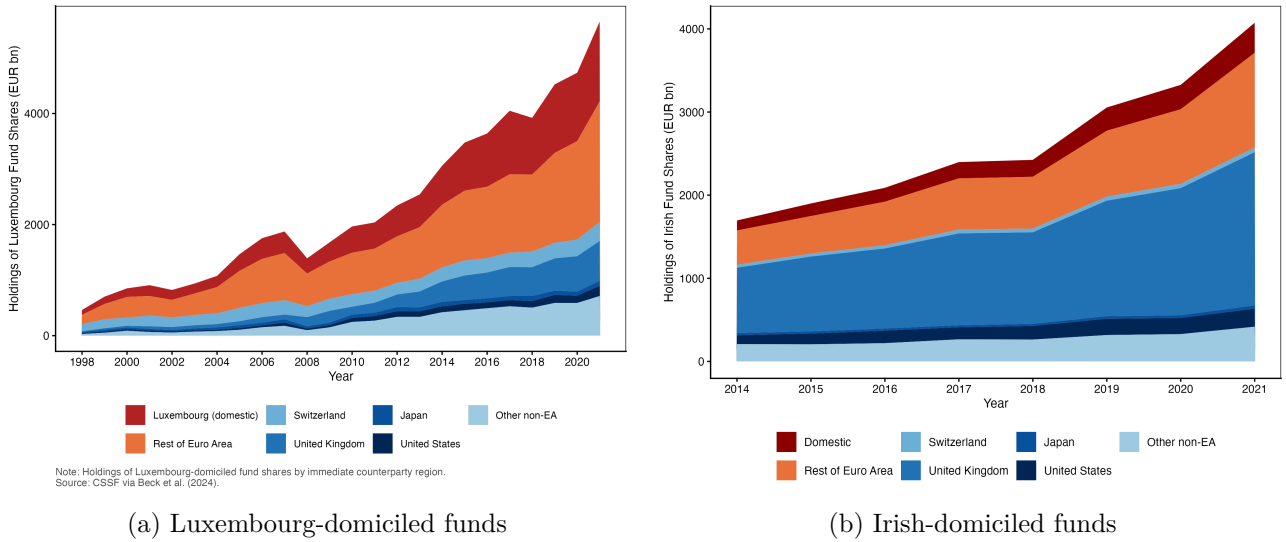


Figure 9: Holdings of fund shares by immediate counterparty region

Note: ‘Domestic’ refers to Luxembourg and Irish investors, respectively. Source: CSSF and Central Bank of Ireland via Beck et al. (2024).

C.3 Appendix C: Corrected Foreign Equity Shares

Figure 10 shows the reported and the corrected foreign equity portfolio share for the EA on the left panel. The right panel displays the same for the cross country average weighted by total equity portfolio. The dotted grey and blue lines show the same excluding Ireland and Luxembourg as it was in Figure 3a. The corrected line for the EA average is higher than for the aggregate but with a similar slope which intuitively makes sense because the foreign equity portfolio is relatively smaller when considering all EA as domestic. Because the data only start in 2014 and the trends are similar, we cannot infer whether this level difference comes from Euro integration or pre-existed simply due to proximity. Furthermore the level difference does not allow for conclusions about the EHB level differences because the second part of equation ?? namely $(1-A)$ is smaller for the EA as for the cross country average. The correction is very large across countries, independent of including Ireland and Luxembourg. Since we saw the rather small magnitude of round tripping which distorts the national EHB, most of this correction presumably comes from re-classification of equity to debt assets if the underlying security of the fund are bonds as described in Beck et al. (March 2024). The small correction for the EA aggregate however is somewhat surprising given the strong decreasing trend in the EA aggregate that initially motivated this analysis. Potentially, there are other erroneous forces driving the strong observed decline for example the stock market capitalisation data.

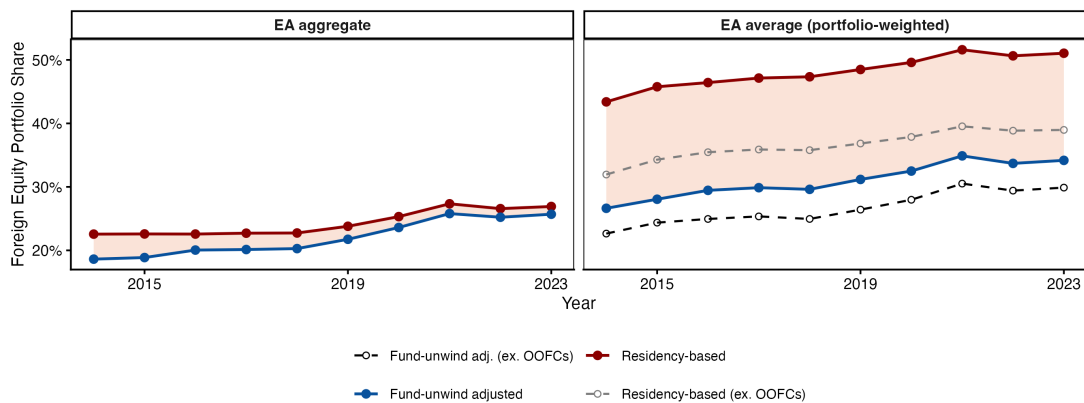
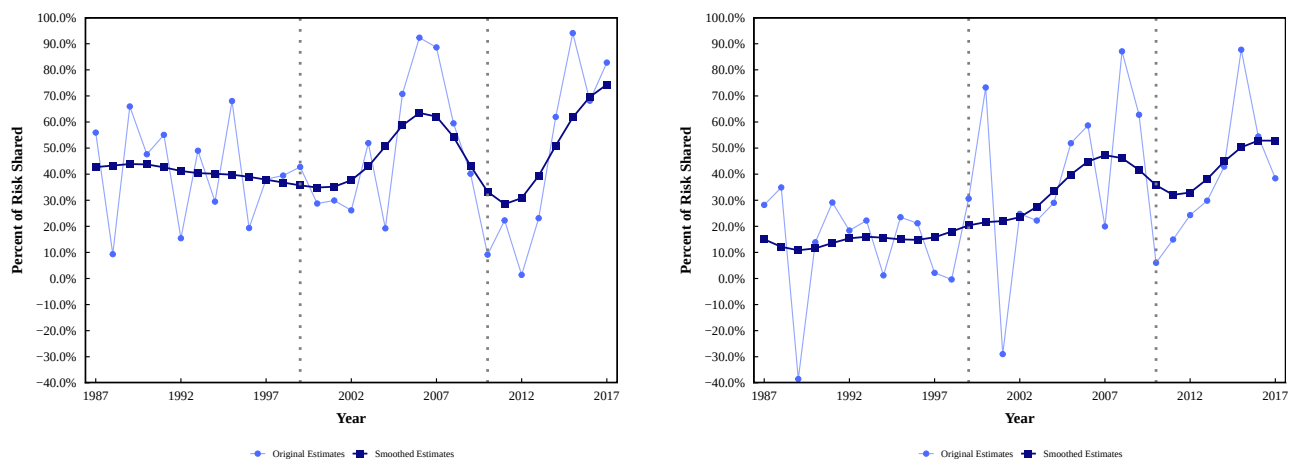


Figure 10: Foreign equity portfolio share: residency-based vs. fund-unwind adjusted

Note: Left panel shows the EA as a single entity; right panel the portfolio-weighted cross-country average. Solid lines include all EA countries; dashed lines exclude Luxembourg, Ireland, and the Netherlands. Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

C.4 Appendix C: Within Group Robustness Check and Panel Risk Sharing Regression

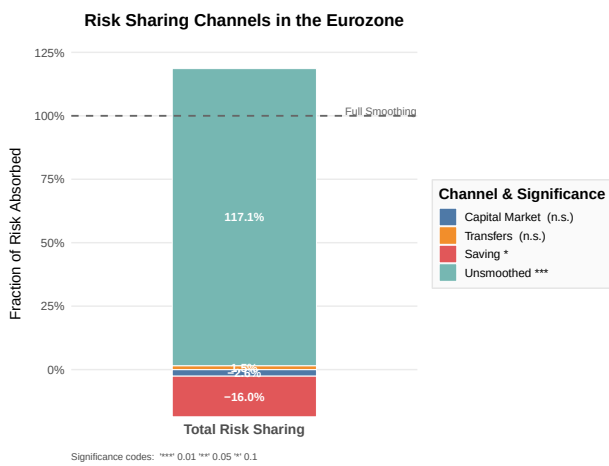


(a) Eurozone only - Consumption

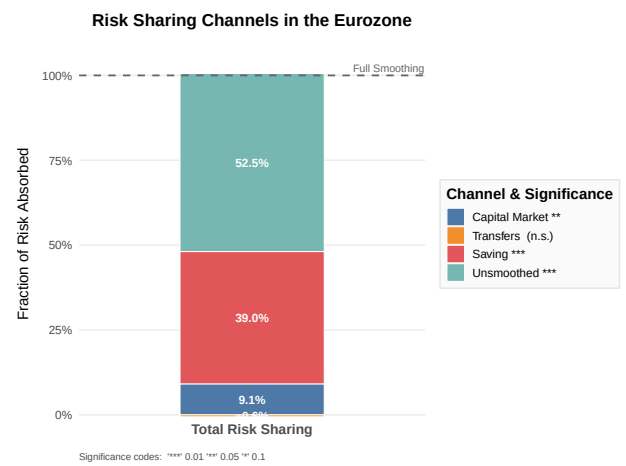
(b) Non-Eurozone - Consumption

Figure 11: Risk Sharing cross-sectional estimates for selected subgroups (Within-group aggregates)

C.5 Appendix C: Corrected Foreign Equity Shares



(a) 1995 – 2007



(b) 2010 – 2023

Figure 12: Risk Sharing Channels (Panel estimation) across different time periods